

AI-Based Interview Preparation Platform

Version: 1.0

Prepared By:

Dhruti Movaliya (24DCS057)

Dhrupal Godhani (24DCS029)

Under Internal Guide :

Mr. Sandeep Maheta

Extrenal Guide :

Mr. Pratham Thacker

Department: Computer Science & Engineering

Institute: DEPSTAR, CHARUSAT

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SOFTWARE REQUIREMENTS SPECIFICATION

AI-Powered Interview Preparation Platform (AIIPP)

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Prepared in accordance with international requirements engineering standards.

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CHAPTER 1 — INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to formally define, describe, and communicate the complete set of functional, non-functional, interface, database, security, and operational requirements of the **AI-Powered Interview Preparation Platform (AIIPP)**. This SRS acts as the primary contractual and technical baseline between all stakeholders — the academic evaluation committee, the software development team, quality assurance personnel, and end users.

Specific purposes:

- Provide a comprehensive, unambiguous, and verifiable description of the System behaviour.
- Establish the scope and boundaries of the platform, clarifying included and excluded features.
- Serve as the reference document for architecture, database design, UI/UX design (Figma), backend (FastAPI), frontend (React.js), and AI integration (Gemini).
- Provide the foundation for verification and validation activities including unit, integration, and user acceptance testing.
- Ensure conformance with IEEE 830-1998 and ISO/IEC/IEEE 29148:2018.
- Present the Final Year Project committee with an industry-grade requirements document.

1.2 Scope

The AI-Powered Interview Preparation Platform is a full-stack, cloud-deployed, AI-driven web application designed to help students, fresh graduates, and working professionals systematically prepare for technical and non-technical job interviews. The platform delivers a suite of intelligent services powered by Google Gemini that automate resume analysis, LinkedIn profile evaluation, project review, AI mock interviewing, personalised roadmap generation, and subject/domain-wise learning support.

1.2.1 Product Name

AI-Powered Interview Preparation Platform (hereafter referred to as **AIIPP** or **the System**).

1.2.2 In-Scope Capabilities

- **User Management** — secure registration, authentication, password recovery, profile management, and session lifecycle using Supabase Auth and JWT.
- **Resume Analyzer** — parsing PDF/DOCX resumes, ATS score, skills extraction, missing-skills detection, grammar/formatting analysis, rating, and downloadable PDF reports.
- **LinkedIn Profile Analyzer** — analysis of public LinkedIn URLs to evaluate headline, About section, experience narrative, skills, keyword density, and profile strength.
- **Project Analyzer** — ingestion of project descriptions and public GitHub repository URLs to detect technologies and rate project quality.
- **Subject-wise Interview Preparation** — DSA, DBMS, OS, CN, OOP, SE, Web with quizzes, AI-generated explanations, and progress tracking.
- **Domain-wise Preparation** — Frontend, Backend, Data Science, AI/ML, Cybersecurity, Cloud, DevOps, Android.
- **AI Mock Interview** — chat-based simulated interviews with configurable difficulty (behavioral, technical, coding).
- **AI Evaluation & Feedback** — post-interview multi-dimensional scoring, strengths, weaknesses, and downloadable report.
- **Personalized Learning Roadmap** — AI-generated weekly plans derived from skill-gap analysis.

- **Dashboard & Progress Tracking** — unified dashboard with charts, statistics, achievements, and notifications.

1.2.3 Out-of-Scope Capabilities (Version 1.0)

- Native mobile applications for Android and iOS.
- Live voice or video AI interviews.
- Real-time integrated coding execution sandbox.
- Direct integration with employer Applicant Tracking Systems.
- Payment and subscription commerce.
- Multilingual content beyond English.

1.2.4 Business Objectives

- Reduce average preparation time through personalized AI-curated content.
- Provide objective, data-driven feedback removing subjectivity of peer mocks.
- Bridge the gap between academic curricula and industry expectations.
- Establish a scalable EdTech product foundation for potential commercialization.

1.3 Intended Audience

This document is intended for the following categories of readers:

Audience	Primary Interest	Recommended Sections
Academic Supervisor & FYP Committee	Standards compliance, completeness	All chapters
Project Development Team	Implementation, modules, architecture	2, 3, 6, 7, 8
UI/UX Designers	Interfaces and interaction flows	3, 4.1, 9
Backend Engineers	Functional requirements, APIs, schema	3, 4.3, 6, 7
Frontend Engineers	UI requirements, external interfaces	3, 4.1, 4.4
Quality Assurance Team	Testable requirements, acceptance criteria	3, 5, 13
DevOps Engineers	Architecture, environment, portability	2.4, 7, 11
Security Auditors	Security and privacy controls	5, 11, 12
End Users (indirect)	Product features	2.2, 2.3

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
AI	Artificial Intelligence.
AIIPP	AI-Powered Interview Preparation Platform.
API	Application Programming Interface.
ATS	Applicant Tracking System.
CRUD	Create, Read, Update, Delete.
CSRF	Cross-Site Request Forgery.
DBMS	Database Management System.
EdTech	Educational Technology.

Term	Definition
ER	Entity-Relationship.
FR	Functional Requirement.
Gemini	Google's generative AI model family used via the Gemini API.
GDPR	General Data Protection Regulation.
HTTPS	HyperText Transfer Protocol Secure.
JSON	JavaScript Object Notation.
JWT	JSON Web Token.
LLM	Large Language Model.
NFR	Non-Functional Requirement.
PII	Personally Identifiable Information.
REST	Representational State Transfer.
RBAC	Role-Based Access Control.
SPA	Single-Page Application.
SRS	Software Requirements Specification.
Supabase	Open-source Backend-as-a-Service providing Postgres, Auth, and Storage.
TC	Test Case.
TLS	Transport Layer Security.
UAT	User Acceptance Testing.
UC	Use Case.
UI/UX	User Interface / User Experience.
UML	Unified Modeling Language.
XSS	Cross-Site Scripting.

1.5 References

- IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
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1.6 Document Overview

This SRS is organised into fifteen chapters and ten appendices. Chapter 1 presents purpose and scope; Chapter 2 gives the overall description; Chapter 3 documents functional module requirements in detail; Chapter 4 covers external interfaces; Chapter 5 lists non-functional requirements; Chapter 6 covers database design; Chapter 7 defines system architecture; Chapters 8–10 provide use cases, activity and sequence descriptions; Chapters 11 and 12 cover security and risk; Chapter 13 defines testing; Chapters 14 and 15 conclude with future work and glossary. Appendices A–J contain reference tables.

CHAPTER 2 — OVERALL DESCRIPTION

2.1 Product Perspective

AIIPP is a **new, self-contained web-based product** that operates independently but relies on external services for its intelligence and infrastructure layers. It is not a replacement for an existing product; however, it competes conceptually with platforms such as InterviewBit, Pramp, and LeetCode by combining resume analysis, mock interviews, and personalized roadmaps into a single AI-driven experience.

The product uses a three-tier client-server architecture augmented by an AI microservice layer:

- **Presentation Tier** — React.js SPA styled with Tailwind CSS, hosted on Vercel.
- **Application Tier** — FastAPI Python service exposing REST endpoints, deployed on Render.
- **Data Tier** — PostgreSQL managed via Supabase; Supabase Storage for file blobs.
- **AI Tier** — Google Gemini API accessed via secure server-to-server calls from FastAPI.

2.1.1 System Interfaces (High-Level)

Interface	Type	Purpose
Browser ↔ Frontend	HTTPS/HTML	User interaction
Frontend ↔ Backend	HTTPS/REST/JSON	Business operations
Backend ↔ Supabase	HTTPS/PostgREST/SQL	Data persistence and auth
Backend ↔ Gemini API	HTTPS/REST	AI generation
Backend ↔ Supabase Storage	HTTPS/S3-like	File storage

2.2 Product Functions

- Account creation, authentication, and profile lifecycle.
- Resume upload, parsing, ATS scoring, and improvement guidance.
- LinkedIn URL analysis with strength scoring.
- Project and GitHub repository quality evaluation.
- Subject-wise structured curriculum with quizzes.
- Domain-wise career-track preparation with roadmaps.
- AI-driven chat-based mock interviews with configurable difficulty.
- Automated multi-dimensional evaluation and downloadable feedback reports.
- Personalized weekly learning plans generated from skill-gap analysis.
- Dashboard with charts, statistics, achievements, and notifications.

2.3 User Classes

2.3.1 Students (Undergraduate / Postgraduate)

Technical Expertise: Beginner to intermediate.

Frequency of Use: Multiple times per week during academic semesters.

Needs: Structured subject-wise preparation, mock interviews, roadmaps.

2.3.2 Fresh Graduates

Technical Expertise: Intermediate.

Frequency of Use: Daily during active job-seeking phase.

Needs: Resume optimization, LinkedIn improvement, targeted mock interviews.

2.3.3 Working Professionals

Technical Expertise: Intermediate to advanced.
Frequency of Use: Occasional, targeted preparation.
Needs: Domain-specific advanced questions, project review, roadmap for career switches.

2.3.4 Administrator

Technical Expertise: Advanced.
Frequency of Use: Daily operational usage.
Needs: Manage users, moderate content, monitor system usage, review AI cost dashboards.

2.4 Operating Environment

2.4.1 Hardware Requirements (Client)

Component	Minimum	Recommended
Processor	Dual-core 1.6 GHz	Quad-core 2.5 GHz
RAM	4 GB	8 GB
Storage	200 MB free	1 GB free
Display	1280×720	1920×1080
Network	2 Mbps	10 Mbps

2.4.2 Software Requirements (Client)

Component	Supported
Operating System	Windows 10/11, macOS 12+, Linux (Ubuntu 20.04+), Android/iOS latest
Browser	Chrome ≥ 110, Firefox ≥ 110, Edge ≥ 110, Safari ≥ 15
JavaScript	Enabled
Cookies	Enabled

2.4.3 Development Tools

Category	Tool
UI/UX Design	Figma
Frontend Framework	React.js 18+
Styling	Tailwind CSS 3+
Backend Framework	FastAPI (Python 3.11+)
Database	PostgreSQL 15 (Supabase-managed)
Authentication	Supabase Auth + JWT
AI	Google Gemini API
Storage	Supabase Storage
Version Control	Git & GitHub
API Testing	Postman
IDE	Visual Studio Code

2.4.4 Deployment Environment

Layer	Provider
Frontend Hosting	Vercel (Edge network, HTTPS enforced)
Backend Hosting	Render (Docker container, autoscaling)
Database & Auth	Supabase (US/EU region)
Storage	Supabase Storage buckets
AI Services	Google Cloud (Gemini API)

2.5 Design Constraints

- Implementation must use the mandated stack (React.js, Tailwind, FastAPI, PostgreSQL/Supabase, Gemini).
- All AI calls routed through the backend; the frontend shall never call Gemini directly.
- All communication over HTTPS with TLS 1.2 or higher.
- Resume file size shall not exceed 5 MB per upload.
- The System shall be responsive across desktop, tablet, and mobile viewports.
- All timestamps shall be stored in UTC.
- Vendor lock-in shall be minimised through standard SQL and open protocols.

2.6 User Documentation

- User Manual (PDF and web).
- In-app help with tooltips and contextual guidance.
- Video tutorials on the platform's help page.
- FAQ section with searchable knowledge base.
- Administrator guide for user and content management.
- API documentation auto-generated via FastAPI's OpenAPI/Swagger UI.

2.7 Assumptions and Dependencies

Assumptions

- Users have stable internet connectivity of at least 2 Mbps.
- Users possess valid email addresses for account creation.
- LinkedIn profiles submitted are public or accessible via paste.
- GitHub repositories submitted are public.
- Users provide truthful data in resumes and profiles.

Dependencies

- Continued availability of the Google Gemini API.
- Continued availability of Supabase services (Auth, DB, Storage).
- Vercel and Render remaining operational.
- Users' browsers supporting modern ECMAScript standards.

2.8 Product Features Summary

#	Feature	Category
F1	User Registration & Authentication	Core
F2	Resume Analyzer	AI

#	Feature	Category
F3	LinkedIn Profile Analyzer	AI
F4	Project & GitHub Analyzer	AI
F5	Subject-wise Preparation	Learning
F6	Domain-wise Preparation	Learning
F7	AI Mock Interview	AI
F8	AI Evaluation & Feedback	AI
F9	Personalized Roadmap	AI
F10	Dashboard & Analytics	Core

CHAPTER 3 — SYSTEM FEATURES

This chapter documents every functional module of the AIIPP System. Each module follows an identical, IEEE-compliant structure covering overview, feature description, priority, actors, preconditions, postconditions, inputs, outputs, functional requirements, basic flow, alternative flow, business rules, and acceptance criteria.

3.1 Module 1 — User Management

3.1 Module Overview

The User Management module governs the complete lifecycle of a user account, including registration, authentication, password recovery, profile maintenance, and session termination. It is the foundation on which every other module depends.

3.1.1 Feature Description

This module allows users to create an account, log in securely, recover forgotten credentials, edit their profile, and log out. Authentication is delegated to Supabase Auth, which issues JWTs consumed by the FastAPI backend. Passwords are hashed with bcrypt.

3.1.2 Priority

High — blocking prerequisite for all other modules.

3.1.3 Actors

Primary: Guest User, Registered User. Secondary: Administrator, Supabase Auth Service, Email Delivery Service.

3.1.4 Preconditions

- User has network connectivity and a modern browser.
- For login, the user has previously registered and verified their email.

3.1.5 Postconditions

- On registration, a users row is created and a verification email is dispatched.
- On login, a JWT is issued and stored securely by the client.
- On logout, the active session token is invalidated.

3.1.6 Inputs

Field	Type	Constraints
Full Name	String	2–80 characters
Email	String	Valid RFC 5322 email, unique
Password	String	Min 8 chars, 1 upper, 1 lower, 1 digit, 1 special
Confirm Password	String	Must match Password
Role	Enum	student / graduate / professional
Profile Picture	File	JPG/PNG, ≤ 2 MB

3.1.7 Outputs

Output	Description
Verification Email	Sent to user's email address
JWT Access Token	Returned on login

Output	Description
User Profile Payload	JSON of profile data
Success/Error Notification	Displayed to user

3.1.8 Functional Requirements

Req ID	Functional Requirement	Priority
FR-UM-01	The system shall allow a guest to register using name, email, and password.	High
FR-UM-02	The system shall validate email format and password strength at both client and server.	High
FR-UM-03	The system shall send an email verification link valid for 24 hours.	High
FR-UM-04	The system shall authenticate registered users via email/password and issue a JWT valid for 60 minutes.	High
FR-UM-05	The system shall provide a Forgot Password flow that emails a secure reset link.	High
FR-UM-06	The system shall allow users to update profile fields (name, role, photo, bio).	Medium
FR-UM-07	The system shall allow users to change their password with old-password confirmation.	High
FR-UM-08	The system shall log out a user by invalidating the token and clearing client storage.	High
FR-UM-09	The system shall lock an account after 5 consecutive failed login attempts for 15 minutes.	High
FR-UM-10	The system shall support role-based access control (user, admin).	High

3.1.9 Basic Flow

1. Guest opens the Registration screen.
2. Guest enters required data and submits.
3. Frontend validates fields and calls POST /auth/register.
4. Backend delegates account creation to Supabase Auth.
5. Verification email is dispatched.
6. User clicks link; email marked verified.
7. User logs in with credentials; JWT returned.
8. Frontend stores token in HttpOnly cookie / secure storage.

3.1.10 Alternative Flow

- A1 – Email already exists: system returns 409; user redirected to login.
- A2 – Weak password: validator rejects submission with guidance.
- A3 – Google OAuth (future): user chooses social login; Supabase OAuth handshake.

3.1.11 Business Rules

- BR-UM-1: Each email may correspond to only one account.
- BR-UM-2: Passwords are never stored in plaintext or logs.
- BR-UM-3: Verification is mandatory before accessing AI features.

3.1.12 Acceptance Criteria

- AC-UM-1: A new user completes registration and receives an email within 30 seconds.
- AC-UM-2: An existing user logs in within 2 seconds under nominal load.
- AC-UM-3: Password reset link expires after single use or 30 minutes.

3.2 Module 2 — Resume Analyzer

3.2 Module Overview

The Resume Analyzer accepts an uploaded resume, parses its content, computes an ATS compatibility score, extracts skills, detects missing skills relative to a target role, and generates recommendations.

3.2.1 Feature Description

Users upload a PDF or DOCX resume. The system extracts text, sends structured prompts to Gemini, and returns a rich analysis including ATS score, skill inventory, gaps, grammar/formatting issues, an overall rating (0–100), and improvement suggestions. A downloadable PDF report is generated.

3.2.2 Priority

High.

3.2.3 Actors

Primary: Authenticated User. Secondary: Gemini API, Supabase Storage, PDF Report Generator.

3.2.4 Preconditions

- User is authenticated and email-verified.
- User possesses a resume file (PDF/DOCX ≤ 5 MB).

3.2.5 Postconditions

- Resume metadata and analysis are stored in resumes and resume_analysis tables.
- Downloadable report accessible from Resume History.

3.2.6 Inputs

Field	Type	Constraints
Resume File	File (PDF/DOCX)	≤ 5 MB
Target Role	String	Optional; e.g., Frontend Developer
Target Industry	String	Optional
Experience Level	Enum	Fresher / 1-3 yrs / 3-5 yrs / 5+ yrs

3.2.7 Outputs

Output	Description
ATS Score	Integer 0–100
Extracted Skills	List of strings
Missing Skills	List with reason
Grammar Issues	List with location
Formatting Issues	List with severity
Overall Rating	Integer 0–100
Suggestions	Prioritised bullet list
PDF Report	Downloadable file

3.2.8 Functional Requirements

Req ID	Functional Requirement	Priority
FR-RA-01	The system shall accept resume uploads in PDF and DOCX formats.	High
FR-RA-02	The system shall reject files exceeding 5 MB.	High
FR-RA-03	The system shall parse text content within 15 seconds for a 3-page resume.	High
FR-RA-04	The system shall compute an ATS score between 0 and 100.	High
FR-RA-05	The system shall extract at least 90% of technical skills present in the document.	Medium
FR-RA-06	The system shall generate at least 5 actionable improvement suggestions per report.	Medium
FR-RA-07	The system shall render a downloadable PDF report.	High
FR-RA-08	The system shall retain resume analysis history per user.	Medium
FR-RA-09	The system shall allow the user to delete an uploaded resume.	High
FR-RA-10	The system shall scan uploaded files for malware signatures before storage.	High

3.2.9 Basic Flow

1. User navigates to Resume Analyzer.
2. User selects file and optional target role.
3. Frontend uploads file to backend endpoint.
4. Backend stores file in Supabase Storage.
5. Backend extracts text via PyPDF2 / python-docx.
6. Backend calls Gemini with structured prompt.
7. Gemini returns JSON analysis.
8. Backend persists results and returns to frontend.
9. Frontend renders analysis dashboard.
10. User optionally downloads PDF report.

3.2.10 Alternative Flow

- A1 – Unsupported format: system returns 415 with message.
- A2 – Empty file or unreadable text: system prompts user to re-upload.
- A3 – Gemini timeout: system retries once, then returns cached fallback rubric.

3.2.11 Business Rules

- BR-RA-1: Analyses are private to the uploading user unless explicitly shared.
- BR-RA-2: Free tier is limited to 3 analyses per day.

3.2.12 Acceptance Criteria

- AC-RA-1: End-to-end analysis completes in ≤ 30 seconds at the 95th percentile.
- AC-RA-2: ATS score reproducibility ± 3 points across two runs of the same file.
- AC-RA-3: PDF report contains all analysis sections.

3.3 Module 3 — LinkedIn Profile Analyzer

3.3 Module Overview

This module analyzes a LinkedIn profile URL and evaluates the headline, About section, work experience, endorsed skills, keyword optimisation, and overall profile strength.

3.3.1 Feature Description

The user provides a public LinkedIn profile URL. The backend fetches the public HTML (or accepts pasted content as fallback), parses relevant sections, and invokes Gemini to score and suggest improvements.

3.3.2 Priority

High.

3.3.3 Actors

Primary: Authenticated User. Secondary: Gemini API, Parser Service.

3.3.4 Preconditions

- User is authenticated.
- Profile is public or user pastes exported content.

3.3.5 Postconditions

- A linkedin_profiles row is created with analysis JSON.

3.3.6 Inputs

Field	Type	Constraints
Profile URL	URL	Must match linkedin.com/in/* pattern
Pasted Content	Text	Optional; for private profiles
Target Role	String	Optional

3.3.7 Outputs

Output	Description
Headline Feedback	Textual analysis
About Feedback	Textual analysis
Experience Feedback	Per-role suggestions
Skills Evaluation	Present and missing skills
Keyword Score	0–100
Profile Strength	0–100
Suggestions	List
PDF Report	Downloadable

3.3.8 Functional Requirements

Req ID	Functional Requirement	Priority
FR-LA-01	The system shall validate the LinkedIn URL format.	High
FR-LA-02	The system shall accept a manually pasted profile block as fallback.	Medium
FR-LA-03	The system shall generate a headline suggestion of ≤ 220 characters.	Medium
FR-LA-04	The system shall compute a profile strength score (0–100).	High
FR-LA-05	The system shall list keyword recommendations relevant to target role.	High

Req ID	Functional Requirement	Priority
FR-LA-06	The system shall retain analysis history.	Medium
FR-LA-07	The system shall render a downloadable PDF report.	High

3.3.9 Basic Flow

1. User opens LinkedIn Analyzer.
2. User pastes URL and optional target role.
3. Backend fetches or reads pasted content.
4. Backend calls Gemini for structured analysis.
5. Analysis stored and returned.
6. User views report and downloads PDF.

3.3.10 Alternative Flow

- A1 – Private profile & no content: prompt user to paste sections.
- A2 – Non-LinkedIn URL: reject with 400 error.

3.3.11 Business Rules

- BR-LA-1: LinkedIn data is retained no longer than 30 days unless user opts in.
- BR-LA-2: Users are warned about LinkedIn's terms of service regarding scraping.

3.3.12 Acceptance Criteria

- AC-LA-1: Report generated in ≤ 25 seconds at P95.
- AC-LA-2: All required output sections are present in the report.

3.4 Module 4 — Project Analyzer

3.4 Module Overview

Analyzes user projects submitted as description text and optional GitHub repository URLs, evaluating technology stack, code quality signals, and descriptive clarity.

3.4.1 Feature Description

The user inputs project title, description, and optional GitHub URL. The backend fetches repo metadata via the GitHub REST API (languages, README, commit frequency, stars, issues) and combines that with a Gemini analysis of the description.

3.4.2 Priority

High.

3.4.3 Actors

Primary: Authenticated User. Secondary: GitHub API, Gemini API.

3.4.4 Preconditions

- User is authenticated.
- GitHub repository (if provided) is public.

3.4.5 Postconditions

- A projects row and project_analysis row are created.

3.4.6 Inputs

Field	Type	Constraints
Project Title	String	3–100 characters
Description	Text	50–3000 characters
GitHub URL	URL	Optional
Role in Project	String	Optional
Duration	String	Optional

3.4.7 Outputs

Output	Description
Detected Technologies	List
Code Quality Feedback	Text
Description Quality Score	0–100
Project Rating	0–100
Suggestions	List

3.4.8 Functional Requirements

Req ID	Functional Requirement	Priority
FR-PA-01	The system shall validate GitHub URLs.	High
FR-PA-02	The system shall fetch repository language breakdown.	High
FR-PA-03	The system shall detect frameworks from package.json, requirements.txt, or pom.xml.	Medium
FR-PA-04	The system shall provide description improvement suggestions.	High
FR-PA-05	The system shall persist analysis history.	Medium

3.4.9 Basic Flow

1. User opens Project Analyzer.
2. Enters title, description, and GitHub URL.
3. Backend fetches GitHub metadata.
4. Gemini analyses description and README.
5. System returns and stores analysis.

3.4.10 Alternative Flow

- A1 – Private repo: system requests README paste.
- A2 – Rate-limited GitHub API: system degrades to description-only analysis.

3.4.11 Business Rules

- BR-PA-1: Analyses use only public data.
- BR-PA-2: Users may delete projects at any time.

3.4.12 Acceptance Criteria

- AC-PA-1: Analysis completes in ≤ 20 seconds at P95.
- AC-PA-2: Detected primary language matches GitHub's declared primary language.

3.5 Module 5 — Subject-wise Interview Preparation

3.5 Module Overview

Organises interview content by academic subject (DSA, DBMS, OS, CN, OOP, SE, Web) with topics, learning materials, AI explanations, practice questions, quizzes, progress tracking, and bookmarks.

3.5.1 Feature Description

The user selects a subject; the system presents a topic tree, curated study material, and interactive quizzes. Gemini generates explanations tailored to the user's proficiency and creates additional practice questions.

3.5.2 Priority

High.

3.5.3 Actors

Primary: Authenticated User. Secondary: Gemini API, Content Repository.

3.5.4 Preconditions

- User is authenticated.
- Subject catalog seeded in the database.

3.5.5 Postconditions

- Progress and bookmark rows updated per topic.

3.5.6 Inputs

Field	Type	Constraints
Subject	Enum	DSA, DBMS, OS, CN, OOP, SE, Web
Topic	String	Selected from list
Quiz Answers	Array	Per quiz session

3.5.7 Outputs

Output	Description
Topic Content	Structured HTML
AI Explanation	Text
Quiz Score	Integer
Progress %	Float
Bookmarks	List

3.5.8 Functional Requirements

Req ID	Functional Requirement	Priority
FR-SP-01	The system shall list at least 7 subjects.	High
FR-SP-02	The system shall list at least 15 topics per subject.	High
FR-SP-03	The system shall auto-grade quizzes and store scores.	High
FR-SP-04	The system shall generate AI explanations on demand.	High
FR-SP-05	The system shall allow bookmarking any topic.	Medium
FR-SP-06	The system shall track and display topic-level progress.	High

3.5.9 Basic Flow

1. User opens Subject Preparation.
2. Selects subject and topic.
3. Reads content and requests AI explanation.
4. Takes quiz; system grades and records progress.

3.5.10 Alternative Flow

- A1 – No content for topic: system falls back to on-demand Gemini generation.

3.5.11 Business Rules

- BR-SP-1: Progress is user-scoped and never shared.
- BR-SP-2: Quiz attempts are capped at 5 per topic per day on the free tier.

3.5.12 Acceptance Criteria

- AC-SP-1: Quiz submission returns result in ≤ 2 seconds.
- AC-SP-2: Progress bar updates in real time.

3.6 Module 6 — Domain-wise Interview Preparation

3.6 Module Overview

Provides career-oriented tracks for Frontend, Backend, Data Science, AI/ML, Cybersecurity, Cloud, DevOps, and Android.

3.6.1 Feature Description

The user selects a domain; the system generates a personalized learning roadmap, curated resources, and practice questions specific to the domain.

3.6.2 Priority

High.

3.6.3 Actors

Primary: Authenticated User. Secondary: Gemini API.

3.6.4 Preconditions

- User is authenticated.

3.6.5 Postconditions

- Domain selection and roadmap saved.

3.6.6 Inputs

Field	Type	Constraints
Domain	Enum	8 supported domains
Current Level	Enum	Beginner / Intermediate / Advanced
Target Timeline	Integer	Weeks 1–52

3.6.7 Outputs

Output	Description
Roadmap	Week-wise steps
Question Bank	Domain-specific

Output	Description
Resources	Links and reading list

3.6.8 Functional Requirements

Req ID	Functional Requirement	Priority
FR-DP-01	The system shall support 8 predefined domains.	High
FR-DP-02	The system shall generate a domain-specific roadmap.	High
FR-DP-03	The system shall present at least 30 practice questions per domain.	Medium
FR-DP-04	The system shall persist the selected domain per user.	Medium

3.6.9 Basic Flow

1. User opens Domain Preparation.
2. Selects domain, level, and timeline.
3. Gemini generates roadmap and question set.
4. User consumes content and marks completion.

3.6.10 Alternative Flow

- A1 – Roadmap regeneration: user may request a new plan; previous archived.

3.6.11 Business Rules

- BR-DP-1: Only one active roadmap per domain per user.

3.6.12 Acceptance Criteria

- AC-DP-1: Roadmap generation ≤ 20 seconds.
- AC-DP-2: Resources open in a new tab.

3.7 Module 7 — AI Mock Interview

3.7 Module Overview

Chat-based mock interview in which Gemini acts as the interviewer, spanning behavioral, technical, and coding questions.

3.7.1 Feature Description

The user configures role, difficulty, and duration; Gemini asks questions turn-by-turn. Answers are stored and used later for evaluation.

3.7.2 Priority

Critical.

3.7.3 Actors

Primary: Authenticated User. Secondary: Gemini API, Session Manager.

3.7.4 Preconditions

- User is authenticated.
- Optional resume analysis available for personalization.

3.7.5 Postconditions

- An interview_sessions row is created; conversation persisted.

3.7.6 Inputs

Field	Type	Constraints
Role	String	e.g., Frontend Developer
Difficulty	Enum	Easy / Medium / Hard
Categories	Multi-select	Behavioral, Technical, Coding
Duration	Integer	15 / 30 / 45 / 60 min
User Answer	Text	Per turn

3.7.7 Outputs

Output	Description
Question	AI-generated
Follow-up	Contextual
Conversation Log	JSON

3.7.8 Functional Requirements

Req ID	Functional Requirement	Priority
FR-MI-01	The system shall allow users to configure role, difficulty, and duration.	High
FR-MI-02	The system shall generate context-aware questions.	High
FR-MI-03	The system shall preserve conversation history within a session.	High
FR-MI-04	The system shall allow the user to end the interview at any time.	High
FR-MI-05	The system shall enforce a maximum session duration.	Medium
FR-MI-06	The system shall automatically move to the next question after user submission.	High

3.7.9 Basic Flow

1. User opens Mock Interview.
2. Configures parameters.
3. System generates first question.
4. User answers; system generates follow-up or next question.
5. Interview ends by time or user action.

3.7.10 Alternative Flow

- A1 – Network drop: session auto-saved and resumable within 24 hours.
- A2 – Gemini error: fallback question bank is used.

3.7.11 Business Rules

- BR-MI-1: Sessions are private.
- BR-MI-2: Free tier limited to 3 mock interviews per week.

3.7.12 Acceptance Criteria

- AC-MI-1: Turn latency ≤ 5 seconds at P95.
- AC-MI-2: Conversation stored verbatim.

3.8 Module 8 — AI Evaluation & Feedback

3.8 Module Overview

Post-interview evaluator computing scores for communication, technical accuracy, and confidence, plus strengths, weaknesses, suggestions, and a composite final score.

3.8.1 Feature Description

When an interview ends, the entire transcript is submitted to Gemini using a structured evaluation prompt returning quantitative scores and qualitative feedback. Results are saved and downloadable as a PDF.

3.8.2 Priority

Critical.

3.8.3 Actors

Primary: Authenticated User. Secondary: Gemini API, PDF Generator.

3.8.4 Preconditions

- An interview session exists and is marked complete.

3.8.5 Postconditions

- An interview_results row is inserted.

3.8.6 Inputs

Field	Type	Constraints
Session ID	UUID	Must belong to user
Evaluation Rubric	Enum	Default / Company-tuned

3.8.7 Outputs

Output	Description
Communication Score	0–100
Technical Score	0–100
Confidence Score	0–100
Strengths	List
Weaknesses	List
Suggestions	List
Final Score	0–100 weighted average
PDF Report	Downloadable

3.8.8 Functional Requirements

Req ID	Functional Requirement	Priority
FR-EV-01	The system shall evaluate every completed session.	High
FR-EV-02	The system shall produce three sub-scores plus a final composite.	High
FR-EV-03	The system shall list at least 3 strengths and 3 weaknesses.	Medium
FR-EV-04	The system shall produce a downloadable PDF report.	High
FR-EV-05	The system shall allow users to compare with previous sessions.	Medium

3.8.9 Basic Flow

1. Interview ends.
2. Transcript sent to Gemini with evaluation prompt.
3. JSON evaluation persisted.
4. User views report; downloads PDF.

3.8.10 Alternative Flow

- A1 – Evaluation failed: system queues retry and notifies user.

3.8.11 Business Rules

- BR-EV-1: Evaluations are immutable once written; new attempts create new rows.

3.8.12 Acceptance Criteria

- AC-EV-1: Evaluation produced within 60 seconds of session end.
- AC-EV-2: PDF renders fully on Chrome, Firefox, Edge, and Safari.

3.9 Module 9 — Personalized Learning Roadmap

3.9 Module Overview

Generates a personalized study plan derived from a skill assessment and gap analysis.

3.9.1 Feature Description

The user completes a short assessment; the system computes gaps against a target role and generates a week-by-week plan including recommended courses, projects, and milestones.

3.9.2 Priority

High.

3.9.3 Actors

Primary: Authenticated User. Secondary: Gemini API.

3.9.4 Preconditions

- User is authenticated.
- Assessment completed.

3.9.5 Postconditions

- A roadmaps row is created.

3.9.6 Inputs

Field	Type	Constraints
Target Role	String	Required
Current Skills	Array	From assessment
Availability	Integer	Hours/week 1–40
Timeline	Integer	4–52 weeks

3.9.7 Outputs

Output	Description
Weekly Plan	Array of week objects
Courses	Recommended

Output	Description
Projects	Recommended
Milestones	Timeline entries

3.9.8 Functional Requirements

Req ID	Functional Requirement	Priority
FR-RM-01	The system shall assess at least 20 skills.	High
FR-RM-02	The system shall identify skill gaps per target role.	High
FR-RM-03	The system shall generate a weekly plan.	High
FR-RM-04	The system shall support editing and regenerating the roadmap.	Medium
FR-RM-05	The system shall track completion of each roadmap item.	High

3.9.9 Basic Flow

1. User launches Roadmap.
2. Takes assessment.
3. System generates plan via Gemini.
4. User works through weekly items and marks progress.

3.9.10 Alternative Flow

- A1 – Regenerate plan: existing roadmap archived; new plan created.

3.9.11 Business Rules

- BR-RM-1: Only one active roadmap per user; historical roadmaps retained.

3.9.12 Acceptance Criteria

- AC-RM-1: Roadmap generation ≤ 25 seconds at P95.
- AC-RM-2: Every recommended item is clickable and trackable.

3.10 Module 10 — Dashboard & Progress Tracking

3.10 Module Overview

The Dashboard aggregates user activity, learning statistics, interview history, resume history, roadmap progress, achievements, and notifications.

3.10.1 Feature Description

The Dashboard presents charts and KPIs summarising platform usage and offers daily goals and notifications.

3.10.2 Priority

High.

3.10.3 Actors

Primary: Authenticated User. Secondary: Notification Service.

3.10.4 Preconditions

- User is authenticated with at least one activity recorded.

3.10.5 Postconditions

- Dashboard state reflects the latest data.

3.10.6 Inputs

Field	Type	Constraints
Date Range	Date pair	Optional
Widget Filter	Multi-select	Optional

3.10.7 Outputs

Output	Description
KPIs	Streak, hours, sessions
Charts	Line / bar / radar
Achievements	Badges
Notifications	List

3.10.8 Functional Requirements

Req ID	Functional Requirement	Priority
FR-DB-01	The system shall show daily goals and streak.	High
FR-DB-02	The system shall display at least 4 charts.	Medium
FR-DB-03	The system shall list the last 10 interviews and resumes.	High
FR-DB-04	The system shall push in-app notifications.	Medium
FR-DB-05	The system shall award achievements upon milestones.	Low

3.10.9 Basic Flow

1. User opens Dashboard.
2. Backend aggregates statistics.
3. Frontend renders charts and lists.
4. User interacts with widgets.

3.10.10 Alternative Flow

- A1 – No data yet: display onboarding tips instead of empty charts.

3.10.11 Business Rules

- BR-DB-1: Statistics are computed nightly and cached for 24 hours with on-demand refresh.

3.10.12 Acceptance Criteria

- AC-DB-1: Dashboard renders in ≤ 3 seconds.
- AC-DB-2: All charts responsive on mobile.

CHAPTER 4 — EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces

Every major screen is documented below with purpose, key elements, and interactions.

4.1.1 Landing Page

Public homepage introducing the product. Elements: hero banner, feature grid, testimonials, CTAs ('Get Started', 'Login'), footer with legal links. Fully responsive; loads in under 2 seconds.

4.1.2 Login / Registration

Elements: email, password, Remember Me, links to Forgot Password and Signup. Inline validation and loading state on submit.

4.1.3 Dashboard

Top nav, side nav, KPI cards (streak, hours, average interview score), charts (line: weekly hours, radar: skills, bar: quiz scores, pie: time distribution), notifications drawer, recent activity list.

4.1.4 Resume Analyzer

Drag-and-drop upload zone, target role selector, Analyze button, results tabs (Overview, ATS, Skills, Grammar, Suggestions), Download PDF button.

4.1.5 LinkedIn Analyzer

URL input, paste-content fallback textarea, target role, results panel with strength gauge, sections list, and Download button.

4.1.6 Project Analyzer

Title, description, GitHub URL fields, Fetch Repo button showing detected languages, and results tabs.

4.1.7 Subject Preparation

Subject grid, topic list, content viewer, AI explanation panel, quiz component with timer and question navigator.

4.1.8 Domain Preparation

Domain cards (Frontend, Backend, DS, AI/ML, Security, Cloud, DevOps, Android), level selector, timeline selector, generated roadmap.

4.1.9 Mock Interview

Setup wizard, chat interface, End Interview button, timer, category badges.

4.1.10 Evaluation Report

Radar chart of scores, strengths/weaknesses cards, suggestions list, Download button, comparison chart with previous sessions.

4.1.11 Roadmap

Timeline of weeks, expandable milestones, checkboxes for tasks, progress bar, regenerate button.

4.1.12 Profile

Avatar upload, name, role, bio, change password, delete account.

4.1.13 Settings

Theme toggle, notification preferences, language (English), data export, account deletion.

4.2 Hardware Interfaces

No specialized hardware interfaces are required. Clients require standard input devices (keyboard, mouse, touchscreen). Audio output is not required in v1. Cloud servers rely on underlying provider hardware abstractions (Render, Vercel, Supabase).

4.3 Software Interfaces

Interface	Version	Purpose
Google Gemini API	v1 (gemini-pro / gemini-flash)	AI generation for all modules
Supabase Postgres	15	Data persistence
Supabase Auth	REST + JWT	Authentication
Supabase Storage	S3-compatible	File storage
FastAPI	≥ 0.110	Backend framework
React	18	Frontend framework
Browser	Modern	Presentation runtime
GitHub REST API	v3	Project analyzer

4.4 Communication Interfaces

Protocol	Purpose
HTTPS (TLS 1.2+)	Transport for all traffic
REST	Backend API architectural style
JSON	Payload serialization
JWT (HS256/RS256)	Auth claims
WebSocket (future)	Real-time mock interview streaming
SMTP	Email delivery via Supabase-provided SMTP

CHAPTER 5 — NON-FUNCTIONAL REQUIREMENTS

All non-functional requirements below are measurable and verifiable.

ID	Category	Requirement	Acceptance
NFR-PERF-01	Performance	Dashboard loads in ≤ 3 seconds for 95% of requests.	P95 latency ≤ 3 s @ 200 concurrent users
NFR-PERF-02	Performance	REST APIs respond in ≤ 500 ms for non-AI requests.	P95 latency ≤ 500 ms
NFR-PERF-03	Performance	AI resume analysis completes in ≤ 30 seconds.	Verified via load test
NFR-SEC-01	Security	All traffic encrypted using TLS 1.2+.	SSL Labs grade $\geq A$
NFR-SEC-02	Security	Passwords hashed with bcrypt cost ≥ 10 .	Config audit
NFR-SEC-03	Security	JWTs expire within 60 minutes.	Config audit
NFR-USE-01	Usability	UI operable without a manual for common tasks.	SUS score ≥ 75
NFR-USE-02	Usability	Text has $\geq 4.5:1$ contrast ratio.	WCAG 2.1 AA compliance
NFR-REL-01	Reliability	MTBF ≥ 720 hours.	Uptime monitoring
NFR-AVL-01	Availability	99.5% monthly uptime.	Uptime dashboard
NFR-MAI-01	Maintainability	Codebase enforces $\geq 80\%$ test coverage.	CI report
NFR-POR-01	Portability	Backend runs on any OCI-compliant container platform.	Deployed on Render + local Docker
NFR-SCA-01	Scalability	Backend handles 500 concurrent users w/ horizontal scaling.	Load-test proof
NFR-ACC-01	Accessibility	WCAG 2.1 AA compliance for all screens.	Axe audit clean
NFR-BAK-01	Backup	Full daily DB backups retained 30 days.	Supabase config
NFR-BAK-02	Recovery	RTO ≤ 4 hours; RPO ≤ 24 hours.	DR runbook test
NFR-LOG-01	Logging	All auth events logged with user ID, IP, timestamp.	Log review
NFR-MON-01	Monitoring	System emits metrics for CPU, memory, latency, errors.	Grafana dashboard
NFR-COM-01	Compliance	GDPR data-subject rights supported.	Legal review
NFR-PRV-01	Privacy	Users can export and delete their data.	Feature demo
NFR-DAT-01	Data Protection	Sensitive fields encrypted at rest.	pgcrypto verified

CHAPTER 6 — DATABASE REQUIREMENTS

6.1 Database Description

PostgreSQL 15 (Supabase-managed) is used as the primary data store. UUIDs are used as primary keys. pgcrypto provides field-level encryption where required. Row-Level Security (RLS) restricts each user's data.

6.2 ER Diagram Description

- users (1) — (N) resumes
- users (1) — (N) linkedin_profiles
- users (1) — (N) projects
- users (1) — (N) interview_sessions (1) — (1) interview_results
- users (1) — (N) roadmaps
- subjects (1) — (N) topics
- users (1) — (N) progress (N) — (1) topics
- users (1) — (N) notifications

6.3 Main Tables

6.3.1 users

Field	Type	Key / Description
id	UUID	PK — Primary user ID
email	VARCHAR(255)	UNIQUE — Login email
password_hash	VARCHAR(255)	Bcrypt hash (managed by Supabase Auth)
full_name	VARCHAR(80)	User full name
role	ENUM	student / graduate / professional / admin
avatar_url	TEXT	Storage URL
bio	TEXT	Free text
is_verified	BOOLEAN	Email verification
created_at	TIMESTAMPTZ	Registration time
updated_at	TIMESTAMPTZ	Last update

6.3.2 resumes

Field	Type	Key / Description
id	UUID	PK
user_id	UUID	FK → users.id
file_url	TEXT	Storage URL
file_name	VARCHAR(255)	
uploaded_at	TIMESTAMPTZ	

6.3.3 resume_analysis

Field	Type	Key / Description
id	UUID	PK
resume_id	UUID	FK → resumes.id
ats_score	INT	0–100
skills_extracted	JSONB	
missing_skills	JSONB	
grammar_issues	JSONB	
suggestions	JSONB	
rating	INT	0–100
created_at	TIMESTAMPTZ	

6.3.4 linkedin_profiles

Field	Type	Key / Description
id	UUID	PK
user_id	UUID	FK → users.id
profile_url	TEXT	
analysis	JSONB	
strength_score	INT	0–100
created_at	TIMESTAMPTZ	

6.3.5 projects

Field	Type	Key / Description
id	UUID	PK
user_id	UUID	FK → users.id
title	VARCHAR(100)	
description	TEXT	
github_url	TEXT	
tech_detected	JSONB	
rating	INT	0–100
created_at	TIMESTAMPTZ	

6.3.6 interview_sessions

Field	Type	Key / Description
id	UUID	PK
user_id	UUID	FK → users.id
role	VARCHAR(100)	
difficulty	ENUM	Easy / Medium / Hard
categories	JSONB	
duration_min	INT	

Field	Type	Key / Description
status	ENUM	active / completed / abandoned
started_at	TIMESTAMPTZ	
ended_at	TIMESTAMPTZ	
transcript	JSONB	

6.3.7 interview_results

Field	Type	Key / Description
id	UUID	PK
session_id	UUID	FK → interview_sessions.id
communication_score	INT	0–100
technical_score	INT	0–100
confidence_score	INT	0–100
final_score	INT	0–100
strengths	JSONB	
weaknesses	JSONB	
suggestions	JSONB	
pdf_url	TEXT	
created_at	TIMESTAMPTZ	

6.3.8 roadmaps

Field	Type	Key / Description
id	UUID	PK
user_id	UUID	FK → users.id
target_role	VARCHAR(100)	
plan	JSONB	
status	ENUM	active / archived
created_at	TIMESTAMPTZ	

6.3.9 subjects

Field	Type	Key / Description
id	UUID	PK
name	VARCHAR(100)	UNIQUE
description	TEXT	

6.3.10 topics

Field	Type	Key / Description
id	UUID	PK
subject_id	UUID	FK → subjects.id
name	VARCHAR(200)	

Field	Type	Key / Description
content	TEXT	

6.3.11 domains

Field	Type	Key / Description
id	UUID	PK
name	VARCHAR(100)	UNIQUE
description	TEXT	

6.3.12 progress

Field	Type	Key / Description
id	UUID	PK
user_id	UUID	FK → users.id
topic_id	UUID	FK → topics.id
status	ENUM	not-started / in-progress / complete
score	INT	0–100
updated_at	TIMESTAMPTZ	

6.3.13 notifications

Field	Type	Key / Description
id	UUID	PK
user_id	UUID	FK → users.id
title	VARCHAR(200)	
body	TEXT	
read	BOOLEAN	
created_at	TIMESTAMPTZ	

6.4 Data Dictionary Notes

- UUIDs are generated by `gen_random_uuid()`.
- JSONB is used for evolving structured data.
- ENUMs are enforced at the database layer.
- All timestamps are stored in UTC.
- RLS policies limit each row to `user_id = auth.uid()` unless the role is admin.

CHAPTER 7 — SYSTEM ARCHITECTURE

7.1 Architecture Overview

The System follows a three-tier client-server architecture augmented by a dedicated AI integration layer. It emphasises stateless services, horizontal scalability, and clear separation of concerns.

7.2 Layer Descriptions

7.2.1 Frontend Layer

- React.js 18 single-page application.
- Tailwind CSS for styling; React Router for navigation.
- Redux Toolkit for state management.
- Axios for HTTP with JWT interceptors.
- Deployed on Vercel with a global CDN.

7.2.2 Backend Layer

- FastAPI (Python 3.11+) with async endpoints.
- Pydantic v2 models for validation.
- SQLAlchemy 2 for ORM (with supabase-py where appropriate).
- Deployed on Render with Dockerfile and horizontal autoscaling.

7.2.3 Database Layer

- PostgreSQL 15 in Supabase (with backups, RLS, extensions).
- Storage buckets for resumes and generated PDFs.

7.2.4 AI Layer

- Google Gemini API accessed server-side.
- Prompt templates versioned in a /prompts directory.
- Response caching using Redis (optional).

7.2.5 External APIs

- GitHub REST API (Project Analyzer).
- Supabase (Auth, Storage).
- SMTP (transactional email).

7.3 Request Flow

- Browser → Vercel edge → React SPA loads.
- React → FastAPI over HTTPS with JWT.
- FastAPI validates token and applies RLS context.
- For AI operations, FastAPI calls Gemini asynchronously.
- Data is persisted in Supabase Postgres and Storage.
- Response returned to client and rendered.

7.4 Deployment Architecture

- Frontend: Vercel (auto-deploy on main).
- Backend: Render (Docker, auto-deploy on main).
- Database: Supabase cloud (multi-AZ).
- CI/CD: GitHub Actions running lint, tests, and build.

7.5 Component Descriptions

Component	Responsibility
Auth Module	Registration, login, tokens, RBAC
File Parser	PDF/DOCX text extraction
Prompt Manager	Templates and versioning
Gemini Client	Rate-limited async client
Report Generator	PDF via WeasyPrint
Cache	Redis for AI-response reuse
Scheduler	Celery/APScheduler for nightly jobs
Notification Service	Email and in-app

CHAPTER 8 — USE CASE SPECIFICATION

The following use cases (UC-01 to UC-12) summarise every module.

UC-01: Register Account

Actor(s): Guest

Description: A new user registers.

Preconditions: Email unused.

Postconditions: User row created; verification email sent.

Main Flow: Fill form → submit → email verification sent.

Alternative Flow: Duplicate email → error.

Exception Flow: Email service down → queued.

UC-02: Login

Actor(s): Registered User

Description: User signs in.

Preconditions: Verified account.

Postconditions: JWT issued.

Main Flow: Enter credentials → JWT.

Alternative Flow: OAuth login.

Exception Flow: Locked account → wait 15 minutes.

UC-03: Analyze Resume

Actor(s): Authenticated User

Description: Upload and analyze resume.

Preconditions: User verified.

Postconditions: Analysis stored.

Main Flow: Upload → parse → analyze → view report.

Alternative Flow: Choose target role for tailored analysis.

Exception Flow: Unreadable file → error.

UC-04: Analyze LinkedIn Profile

Actor(s): Authenticated User

Description: Provide URL, receive analysis.

Preconditions: User authenticated.

Postconditions: Analysis stored.

Main Flow: URL → fetch → Gemini analysis → report.

Alternative Flow: Paste content manually.

Exception Flow: Invalid URL → reject.

UC-05: Analyze Project

Actor(s): Authenticated User

Description: Analyze project and optional GitHub repo.

Preconditions: User authenticated.

Postconditions: Analysis stored.

Main Flow: Enter details → optional GitHub → analyze → view rating.

Alternative Flow: No GitHub → text-only analysis.

Exception Flow: GitHub rate limit → degrade.

UC-06: Subject Preparation

Actor(s): Authenticated User

Description: Learn subject topics and take quizzes.

Preconditions: Subject catalog seeded.

Postconditions: Progress updated.

Main Flow: Select subject → view topic → take quiz.

Alternative Flow: Bookmark topics.

Exception Flow: Quiz submission fails → retry.

UC-07: Domain Preparation

Actor(s): Authenticated User

Description: Follow domain-specific track.

Preconditions: User authenticated.

Postconditions: Roadmap stored.

Main Flow: Select domain → generate roadmap → practice.

Alternative Flow: Regenerate roadmap.

Exception Flow: Gemini timeout → fallback template.

UC-08: Conduct Mock Interview

Actor(s): Authenticated User

Description: Take an AI-driven mock interview.

Preconditions: User authenticated.

Postconditions: Session stored.

Main Flow: Setup → chat → end.

Alternative Flow: Pause and resume.

Exception Flow: Session lost → auto-recover.

UC-09: View Evaluation Report

Actor(s): Authenticated User

Description: See post-interview evaluation.

Preconditions: Completed session.

Postconditions: Evaluation stored.

Main Flow: Session end → evaluation → view/download.

Alternative Flow: Compare with previous.

Exception Flow: Evaluation failure → retry.

UC-10: Generate Roadmap

Actor(s): Authenticated User

Description: Get personalized weekly plan.

Preconditions: Assessment complete.

Postconditions: Roadmap stored.

Main Flow: Assessment → generate → track.

Alternative Flow: Regenerate.

Exception Flow: Gemini error → retry.

UC-11: Dashboard

Actor(s): Authenticated User

Description: View KPIs and history.

Preconditions: User has activity.

Postconditions: Dashboard rendered.

Main Flow: Login → dashboard renders KPIs and charts.

Alternative Flow: Filter date range.

Exception Flow: No data → onboarding tips.

UC-12: Administer Users

Actor(s): Administrator

Description: Manage users and content.

Preconditions: Admin privileges.

Postconditions: Audit log written.

Main Flow: View list → suspend/reactivate → audit logs.

Alternative Flow: Bulk export.

Exception Flow: Permission failure → 403.

CHAPTER 9 — ACTIVITY DIAGRAMS

Each activity flow is described textually and is suitable for direct conversion to a UML activity diagram.

AD-01 Registration

Start → Enter Details → Validate → [Valid?] Yes → Create User → Send Email → End; No → Show Errors → back to Enter Details.

AD-02 Login

Start → Enter Credentials → Validate → Issue JWT → Load Dashboard → End.

AD-03 Resume Analysis

Start → Upload File → Validate Format → Store in Bucket → Extract Text → Call Gemini → Persist Analysis → Show Report → End.

AD-04 LinkedIn Analysis

Start → Input URL → Validate → Fetch or Paste Content → Call Gemini → Persist → Display → End.

AD-05 Project Analysis

Start → Input Details → [GitHub URL?] Yes → Fetch Repo Metadata → Merge with Description; No → Description only → Call Gemini → Store → Display → End.

AD-06 Subject Preparation

Start → Select Subject → Choose Topic → View Content → [Quiz?] Yes → Take Quiz → Grade → Update Progress; No → Bookmark → End.

AD-07 Domain Preparation

Start → Select Domain → Enter Level/Timeline → Generate Roadmap → Show → End.

AD-08 Mock Interview

Start → Configure → Generate Q1 → Loop {Show Question → Receive Answer → Next Question} until End/Timeout → Persist → End.

AD-09 Evaluation

Start → Retrieve Transcript → Call Gemini → Compute Scores → Persist → Generate PDF → End.

AD-10 Roadmap

Start → Assessment → Gap Analysis → Generate Plan → Persist → Display → Track Progress → End.

AD-11 Dashboard

Start → Fetch KPIs → Fetch Charts Data → Fetch Notifications → Render → End.

CHAPTER 10 — SEQUENCE DIAGRAMS

Textual sequence descriptions for direct UML conversion.

SD-01 Registration

1. Browser → Frontend: Submit form
2. Frontend → Backend: POST /auth/register
3. Backend → Supabase Auth: create user
4. Supabase → Backend: 201 + JWT
5. Backend → SMTP: send verification
6. Backend → Frontend: 201
7. Frontend → User: Success screen

SD-02 Login

1. Frontend → Backend: POST /auth/login
2. Backend → Supabase: verify credentials
3. Supabase → Backend: JWT
4. Backend → Frontend: 200 + JWT
5. Frontend → Browser: Store token
6. Frontend → Backend: GET /profile (Auth: Bearer)
7. Backend → Frontend: Profile JSON

SD-03 Resume Analysis

1. Frontend → Backend: POST /resume (file)
2. Backend → Supabase Storage: upload
3. Storage → Backend: URL
4. Backend → Parser: extract text
5. Backend → Gemini: analyze prompt
6. Gemini → Backend: JSON
7. Backend → DB: insert resume + analysis
8. Backend → Frontend: payload
9. Frontend → User: Report UI

SD-04 LinkedIn Analysis

1. Frontend → Backend: POST /linkedin
2. Backend → LinkedIn/Parser: fetch content
3. Backend → Gemini: prompt
4. Gemini → Backend: JSON
5. Backend → DB: insert
6. Backend → Frontend: payload

SD-05 Project Analysis

1. Frontend → Backend: POST /project

2. Backend → GitHub: GET repo
3. GitHub → Backend: metadata
4. Backend → Gemini: prompt
5. Gemini → Backend: JSON
6. Backend → DB: insert
7. Backend → Frontend: payload

SD-06 Subject Prep

1. Frontend → Backend: GET /subjects/{id}/topics
2. Backend → DB: query
3. DB → Backend: rows
4. Frontend → Backend: POST /explain {topic}
5. Backend → Gemini: explanation
6. Gemini → Backend: text
7. Backend → Frontend: text

SD-07 Domain Prep

1. Frontend → Backend: POST /domain/roadmap
2. Backend → Gemini: plan prompt
3. Gemini → Backend: JSON
4. Backend → DB: insert roadmap
5. Backend → Frontend: plan

SD-08 Mock Interview

1. Frontend → Backend: POST /interview/start
2. Backend → DB: create session
3. Backend → Gemini: Q1
4. Loop: Frontend → Backend: POST /interview/{id}/answer → Backend → Gemini: next question → Backend → Frontend: question
5. Frontend → Backend: POST /interview/{id}/end
6. Backend → DB: update session

SD-09 Evaluation

1. Backend (post-end) → Gemini: evaluate transcript
2. Gemini → Backend: scores
3. Backend → PDF Service: render
4. PDF Service → Storage: upload
5. Backend → DB: insert result
6. Backend → Frontend: result payload

SD-10 Roadmap

1. Frontend → Backend: POST /roadmap/generate

2. Backend → Gemini: prompt
3. Gemini → Backend: JSON plan
4. Backend → DB: insert
5. Backend → Frontend: plan

SD-11 Dashboard

1. Frontend → Backend: GET /dashboard
2. Backend → DB: aggregate queries
3. DB → Backend: rows
4. Backend → Frontend: KPI + charts JSON

CHAPTER 11 — SECURITY REQUIREMENTS

Category	Requirement
Authentication	Supabase Auth with email verification; MFA optional (future).
Authorization	RBAC with user and admin roles; RLS policies in Postgres.
Password Hashing	bcrypt with cost ≥ 10 managed by Supabase Auth.
JWT	Signed tokens, short lifetime (60 min); refresh via secure endpoint.
Role-Based Access Control	Route-level checks in FastAPI dependency injection.
Input Validation	Pydantic schemas server-side; Yup/Zod client-side.
SQL Injection Prevention	Parameterized queries via ORM; no dynamic string SQL.
XSS Protection	Escape output, sanitize HTML, Content-Security-Policy header.
CSRF Protection	SameSite=Strict cookies; state tokens for form submissions.
HTTPS	Enforced end-to-end with HSTS.
Rate Limiting	100 req/min/user global; stricter caps on AI endpoints.
Secure File Upload	MIME type check, magic-byte inspection, malware scan, 5 MB cap.
Data Encryption	TLS in transit; pgcrypto for sensitive fields; Storage encryption at rest.
Audit Logs	Login, admin actions, data exports logged with actor and IP.
Session Management	Auto-logout after 30 minutes of inactivity.
Privacy Protection	GDPR-compliant data export and delete endpoints.

CHAPTER 12 — RISK ANALYSIS

Risk ID	Description	Probability	Impact	Mitigation
RISK-01	Gemini API outage	Medium	High	Fallback question bank; retry with exponential backoff.
RISK-02	Gemini cost overrun	Medium	High	Rate limits, per-user quotas, and caching.
RISK-03	Data breach of PII	Low	Critical	Encryption, RLS, audits, and minimising PII.
RISK-04	Supabase downtime	Low	High	Nightly backups and a documented DR plan.
RISK-05	Malicious file uploads	Medium	High	Antivirus scan, MIME validation, sandboxed parsing.
RISK-06	LinkedIn ToS conflict	Medium	Medium	Provide manual paste option and disclaimer.
RISK-07	GitHub rate limiting	Medium	Medium	Authenticated tokens and exponential backoff.
RISK-08	Scope creep	High	Medium	Signed baseline SRS and change control.
RISK-09	Schedule slip	Medium	Medium	Weekly checkpoints and MVP-first strategy.
RISK-10	Team knowledge gaps	Medium	Medium	Training sprints and pair programming.
RISK-11	XSS / SQLi vulnerabilities	Low	High	Static analysis and mandatory code review.
RISK-12	Vendor lock-in	Medium	Medium	Abstract vendor calls behind interfaces.
RISK-13	Inaccurate AI feedback	High	Medium	Prompt evaluation, rubric tuning, user ratings.
RISK-14	Browser compatibility	Low	Medium	Cross-browser testing matrix.
RISK-15	Poor onboarding / UX	Medium	Medium	Usability testing and iterative design.

CHAPTER 13 — TESTING REQUIREMENTS

13.1 Testing Levels

Level	Description
Unit Testing	Function/component tests using pytest and Jest.
Integration Testing	API + DB tests using test containers.
System Testing	End-to-end tests via Playwright.
Acceptance Testing	Stakeholder scenarios; checklist per module.
Performance Testing	k6 or Locust load tests.
Security Testing	OWASP ZAP scans and manual review.
Usability Testing	Moderated sessions and SUS surveys.
Compatibility Testing	Cross-browser and device matrix.
Regression Testing	Full suite executed on every release candidate.

13.2 Representative Test Cases

TC ID	Scenario	Steps	Expected
TC-UM-01	Successful registration	Fill form → submit	201 + email sent
TC-UM-02	Duplicate email	Register with existing email	409 Conflict
TC-UM-03	Weak password	Password abc123	Client validation error
TC-RA-01	Valid PDF upload	Upload 2-page PDF	Analysis returned ≤ 30 s
TC-RA-02	Oversize file	Upload 8 MB PDF	Rejected with 413
TC-LA-01	Valid LinkedIn URL	Submit valid URL	Analysis returned
TC-LA-02	Invalid URL	Submit google.com	400 Bad Request
TC-PA-01	Valid GitHub repo	Submit public repo	Languages detected
TC-MI-01	Start interview	Configure and start	Q1 shown
TC-MI-02	End interview	Click End	Session marked complete
TC-EV-01	Evaluate session	Complete interview	Scores + PDF generated
TC-RM-01	Generate roadmap	Complete assessment	Weekly plan shown
TC-DB-01	Dashboard render	Login and open dashboard	Charts + KPIs shown
TC-SEC-01	SQL injection attempt	Send ' OR 1=1 -- to login	Rejected safely
TC-SEC-02	Rate limit	Send > 100 req/min	429 Too Many Requests

CHAPTER 14 — FUTURE ENHANCEMENTS

- **Voice-based AI Interviews** — full-duplex speech interviews using speech-to-text and text-to-speech.
- **Video Interviews** — webcam-based sessions with body-language cues.
- **Real-Time Coding Assessment** — integrated code editor and sandboxed execution engine.
- **Company-specific Interview Preparation** — tailored question banks and rubrics per company.
- **AI Career Mentor** — persistent, long-term career coaching agent.
- **Job Recommendation System** — match user profile to available openings.
- **Mobile Application** — native Android and iOS clients.
- **Multilingual Support** — content and UI in additional languages.
- **Gamification** — XP, leaderboards, and streak rewards.
- **Recruiter Dashboard** — companies can post assessments and view candidate reports.
- **Analytics** — advanced aggregate analytics for institutions and administrators.

CHAPTER 15 — GLOSSARY

Term	Definition
ATS	Applicant Tracking System that filters resumes based on keywords and layout.
Bcrypt	Password-hashing function designed for slow, secure hashing.
Backend	Server-side application logic and API layer.
Container	Lightweight, isolated process bundle (Docker/OCI).
CI/CD	Continuous Integration and Continuous Delivery.
Frontend	Browser-side user interface layer.
Gemini	Google's family of multimodal generative AI models.
JWT	JSON Web Token — self-contained authentication claims.
ORM	Object-Relational Mapping between code and database rows.
Prompt	Input text sent to an LLM to elicit a response.
RLS	Row-Level Security — Postgres access rules per row.
Roadmap	Weekly personalized learning plan produced by the System.
Session	A single interview or authenticated user session.
Supabase	BaaS providing Postgres, Auth, Storage, and Realtime.
TLS	Transport Layer Security.
Webhook	HTTP callback triggered by an event.

APPENDICES

Appendix A — Abbreviations

Abbrev.	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
ATS	Applicant Tracking System
FR	Functional Requirement
JWT	JSON Web Token
LLM	Large Language Model
NFR	Non-Functional Requirement
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
SPA	Single-Page Application
SRS	Software Requirements Specification
TLS	Transport Layer Security
UC	Use Case
UI/UX	User Interface / User Experience

Appendix B — References

- IEEE Std 830-1998.
- ISO/IEC/IEEE 29148:2018.
- Google Gemini API Documentation.
- Supabase Documentation.
- FastAPI Official Documentation.
- React Official Documentation.
- Tailwind CSS Documentation.
- OWASP Top Ten.
- GDPR — Regulation (EU) 2016/679.
- Sommerville, I. Software Engineering, 10th Ed.
- Pressman, R. Software Engineering: A Practitioner's Approach, 8th Ed.

Appendix C — Technology Stack Summary

Layer	Technology
UI/UX Design	Figma
Frontend	React.js 18 + Tailwind CSS 3
Backend	FastAPI (Python 3.11+)
Database	PostgreSQL 15 (Supabase)
Auth	Supabase Auth + JWT
AI	Google Gemini API

Layer	Technology
Storage	Supabase Storage
Version Control	Git & GitHub
API Testing	Postman
Frontend Hosting	Vercel
Backend Hosting	Render
IDE	Visual Studio Code

Appendix D — API Summary

Endpoint	Method	Purpose
/auth/register	POST	Register a new user
/auth/login	POST	Authenticate and issue JWT
/auth/forgot-password	POST	Send password reset link
/auth/reset-password	POST	Reset password with token
/profile	GET/PUT	Retrieve or update user profile
/resume	POST	Upload and analyze resume
/resume/{id}	GET/DELETE	Retrieve or delete resume
/linkedin	POST	Analyze LinkedIn profile
/project	POST	Analyze project & GitHub repo
/subjects	GET	List subjects and topics
/domain/roadmap	POST	Generate domain roadmap
/interview/start	POST	Start a mock interview
/interview/{id}/answer	POST	Submit an interview answer
/interview/{id}/end	POST	End interview session
/evaluation/{session_id}	GET	Retrieve evaluation report
/roadmap/generate	POST	Generate personalized roadmap
/dashboard	GET	Aggregate KPIs and charts
/notifications	GET/PATCH	List or mark notifications

Appendix E — Screen List

- Landing Page
- Login
- Registration
- Forgot Password
- Dashboard
- Resume Analyzer
- LinkedIn Analyzer
- Project Analyzer
- Subject Preparation

- Domain Preparation
- Mock Interview
- Evaluation Report
- Roadmap
- Profile
- Settings
- Notifications
- Admin Console

Appendix F — Database Schema Summary

Table	Purpose
users	Account and profile data
resumes	Uploaded resume files
resume_analysis	AI analysis of resumes
linkedin_profiles	LinkedIn analyses
projects	User project entries
interview_sessions	Mock interview sessions
interview_results	Post-interview evaluations
roadmaps	Personalized learning plans
subjects / topics	Curriculum content
domains	Career-domain reference data
progress	Per-topic user progress
notifications	In-app notifications

Appendix G — Deployment Summary

Layer	Service	Notes
Frontend	Vercel	Auto-deploy from main; global CDN; HTTPS
Backend	Render	Docker container; autoscaling
Database	Supabase Postgres	Managed backups + PITR
Storage	Supabase Storage	Buckets: resumes, reports
AI	Google Gemini API	Server-to-server calls only
CI/CD	GitHub Actions	Lint, tests, build, deploy

Appendix H — Assumptions

- Users have stable internet connectivity.
- Users provide truthful data.
- LinkedIn and GitHub content submitted is public or pasted.
- Gemini and Supabase remain available and within cost limits.

Appendix I — Constraints

- Mandatory stack: React.js, Tailwind, FastAPI, Supabase, Gemini.

- Resume upload maximum 5 MB.
- HTTPS with TLS 1.2+ enforced.
- All timestamps stored in UTC.
- AI calls proxied through backend only.

Appendix J — Revision History

Version	Date	Author	Change Summary
0.1	10 May 2026	Project Team	Initial draft
0.5	20 June 2026	Project Team	Modules 1–5 completed
0.9	15 July 2026	Project Team	All chapters drafted; review
1.0	01 August 2026	Project Team	Baseline approved

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